INCH-POUND
MIL-PRF-3098/61K
w/AMENDMENT 1
16 September 2019
SUPERSEDING
MIL-PRF-3098/61K
7 January 2010

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR84/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

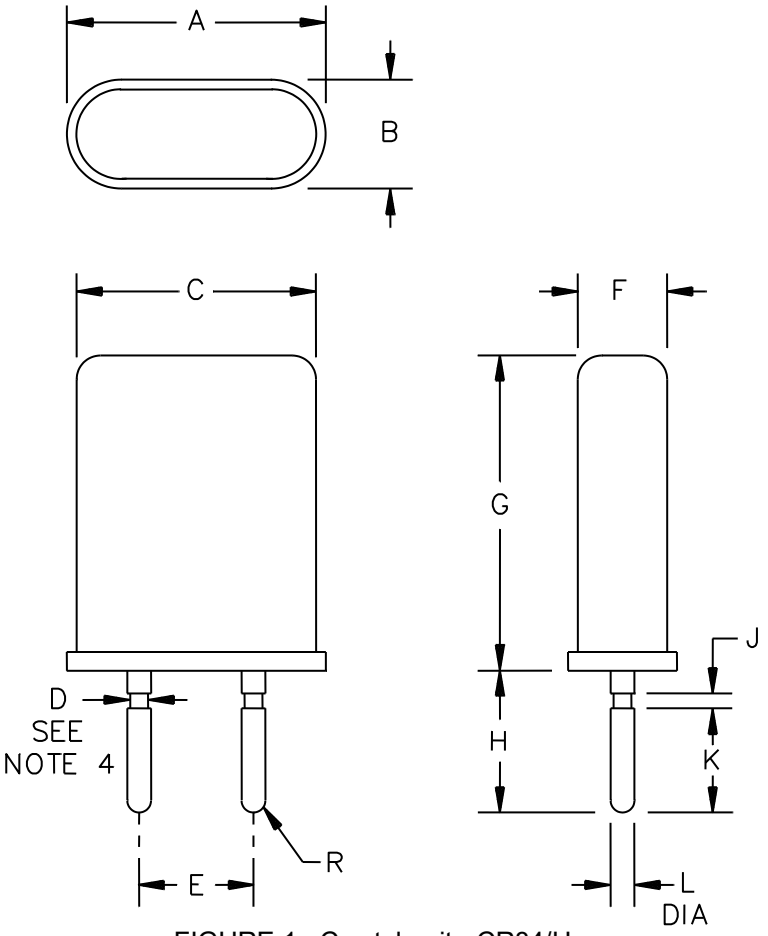



FIGURE 1. Crystal unit - CR84/U.



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| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .435 | --- | 11.05 |
| B | --- | .183 | --- | 4.65 |
| C | --- | .402 | --- | 10.21 |
| D | .027 | .032 | 0.69 | 0.81 |
| E | .184 | .200 | 4.67 | 5.08 |
| F | --- | .150 | --- | 3.81 |
| G | --- | .530 | --- | 13.46 |
| H | .223 | .248 | 5.66 | 6.30 |
| J | .015 | .025 | 0.38 | 0.64 |
| K | .165 | .175 | 4.19 | 4.45 |
| L | .038 | .042 | 0.97 | 1.07 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. Pin undercut may be omitted.

FIGURE 1. Crystal unit - CR84/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17.0 MHz to 61.0 MHz, inclusive.

Frequency tolerances:

Primary operating temperature range: ±20 parts per million (ppm).

Room temperature range: ±70 ppm.

Frequency stability: ±5 ppm

Equivalent resistance: 40 ohms, maximum.

Drive level: 1.0 mW maximum.

Resonance: Series.

Mode of oscillation: Third mechanical overtone.

Reference temperature: +85°C ±1°C.

Operating temperature ranges:
Operable: -55°C to +80°C, and +90°C to +105°C, inclusive.



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Operating: +80°C to +90°C, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: MIL-STD-202-204, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): CR84/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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Custodians:
Army - CR
Navy - EC
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Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2019-021)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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